

A New Generation of USD/INR Prediction Systems: GDELT Sentiment, Variational Decomposition, Multi-Persona LLMs, Market Memory, and Topology-Enabled Predictions from Chaos

Dhruv Kumar^{*} Amrit Lahari[†] Suyash Handa[‡] Shrey Bansal[§]

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Abstract

USD/INR forecasting is difficult because the Indian foreign exchange market is a managed-float system shaped by RBI intervention, imported energy shocks, dollar cycles, geopolitical events, and delayed rather than immediate news absorption. This paper consolidates the full research program embodied in the repository: Phase-A/Phase-B/Phase-C VMD and GDELT pipelines, EGARCH and hybrid volatility models, multi-model ensemble forecasting, regime-switching Monte Carlo simulation, the debiased Gemini V4 multi-persona architecture, the newer market-memory-first **final phase** stack, and the TEPC topology-chaos branch. Across the saved repo artifacts, raw aggregate signals are weak in isolation (Goldstein vs. IMF_3 correlation = 0.010; US average tone vs. USD/INR correlation = 0.164), but structured feature engineering recovers delayed signal: **Tone_Economy** reaches a statistically significant correlation of -0.147 with IMF_3 , meaningful-pattern analysis finds peak lags of 9 days for Goldstein and 12 days for tone, and the Phase-C danger-day classifier achieves 40% precision with 52% recall. In the historical Gemini track, V4 improves over V3 from 0.274% to 0.270% mean hybrid absolute error and from 61.3% to 64.2% directional accuracy under a sign-of-change evaluation, while January 2026 live predictions average 0.322% absolute error. In the **final phase** walk-forward backtest, the memory-only ablation is strongest over 100 test days (MAE price 0.242, directional accuracy 61%), indicating that compact shared state can outperform heavier feature blends. In the strict raw-GDELT TEPC run, **tepc_full** leads on macro-F1 (0.337) with 77.5% breakout accuracy, while persona blending reduces return MAE to 0.002695 and live 100-day runs show chaos features as the strongest classifier. The repository-wide conclusion is that USD/INR rewards regime detection, volatility classification, shared-state compression, and topology-aware modeling far more than naive direct price regression or heavyweight deep sequence models.

Keywords: USD/INR, GDELT, VMD, EGARCH, Monte Carlo, LLM Forecasting, Market Memory, TEPC, Chaos Modeling, Exchange Rate Prediction

1 Introduction

Forecasting exchange rates remains one of the hardest problems in empirical finance. The central puzzle established by [Meese & Rogoff \(1983\)](#) still holds in modern settings: out-of-sample currency forecasting is notoriously brittle, and many structurally motivated models fail

^{*}dhruv.kumar@pilani.bits-pilani.ac.in

[†]f20230442@pilani.bits-pilani.ac.in

[‡]f20230029@pilani.bits-pilani.ac.in

[§]f20230649@pilani.bits-pilani.ac.in

to beat simple benchmarks. This challenge is especially acute for USD/INR, where a managed-float regime, cross-asset dependence on oil and the dollar, event-driven sentiment shocks, and intermittent RBI intervention complicate the mapping from public information to next-day price movement. In such a market, a model can fit history extremely well while remaining economically useless in forward tests.

At the same time, the informational environment has changed. Global news databases such as GDELT provide dense coverage of political, economic, and geopolitical events in near real time (Leetaru & Schrod, 2013). Financial modeling has also broadened from classical volatility models (Bollerslev, 1986; Nelson, 1991) toward boosted trees, neural sequence models, and more recently large language models (LLMs) and multi-agent reasoning systems (Chen & Guestrin, 2016; Sezer et al., 2020; Lopez-Lira & Tang, 2023; Xie et al., 2023; Li et al., 2024). The repository studied in this paper is best understood as a research laboratory built around that shift: it starts from raw GDELT sentiment and macro data, progresses through VMD-based decomposition and danger-day classification, then evolves into LLM-assisted forecasting, memory-first hybrid modeling, and finally topology- and chaos-driven network forecasting.

This paper does not present a single homogeneous benchmark in which every model is evaluated under one frozen protocol. Instead, it documents and synthesizes the full research program captured by the repository. That distinction matters. The project contains multiple branches developed over time for different forecasting objectives: volatility classification, next-day point prediction, probabilistic scenario forecasting, walk-forward ablations under information embargo, and topology-aware breakout modeling. The scientific value lies in comparing these branches as complementary answers to the same core question: *what kinds of signals and model structures actually survive contact with USD/INR?*

The central conclusion that emerges is pragmatic. USD/INR is not a market in which arbitrarily complex models automatically help. Raw aggregate sentiment is too weak; direct price regression is unstable; deep sequence models can fail spectacularly; and naive feature augmentation can worsen performance. Signal appears only after decomposition, thematic filtering, lag engineering, state compression, or network-level modeling. In other words, the market problem is not simply “predict tomorrow’s price”, but rather *identify which layer of the market is predictable*: volatility, directional stress, shared market state, or synchronized macro-network regimes.

1.1 Contributions

Our main contributions are as follows:

- **Repository-wide synthesis of a multi-stage research program.** We consolidate the phase-based VMD/GDELT experiments, GARCH and ensemble branches, Gemini persona systems, the newer market-memory-first stack, and the TEPC topology-chaos branch into one coherent paper.
- **A transparent account of where signal actually appears in USD/INR.** We show that raw aggregate Goldstein signal is nearly useless on its own, while thematic filtering, lag analysis, and structured state representations recover delayed and economically interpretable predictive structure.
- **Evidence that off-the-shelf direct forecasting is poorly matched to this market.** ARIMA, SARIMA, GRU, LSTM, Attention, and several news-augmented regressors all underperform simple benchmarks or produce negative R^2 , whereas volatility classification, market memory, and topology-aware models remain useful.
- **A detailed empirical bridge from LLM prompting to stateful and chaos-aware forecasting.** We document how the repository evolves from Gemini V4 debiased persona systems

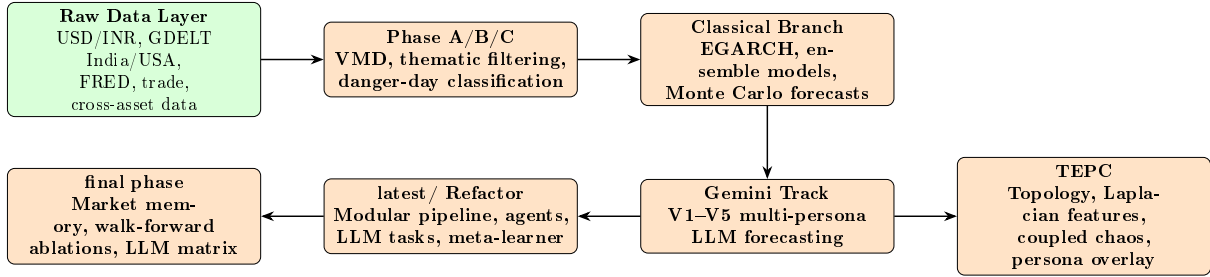


Figure 1: Repository evolution as a research program rather than a single monolithic model. Early signal extraction work (Phase A/B/C) feeds the quantitative branches; Gemini V1–V5 introduces debiased multi-persona forecasting; **latest/** acts as a modular bridge; **final phase** formalizes market memory and walk-forward ablations; TEPC extends the program into dynamic topology and chaos-aware forecasting.

to the **final phase** memory-first framework and finally to TEPC, where graph topology and coupled Lorenz dynamics become first-class features.

2 Related Work

2.1 Exchange-Rate Forecasting and Volatility Modeling

Exchange-rate predictability has long been constrained by the Meese–Rogoff result that structural models struggle to outperform naive random-walk baselines out of sample (Meese & Rogoff, 1983). Later syntheses emphasized that no single specification consistently dominates across periods and regimes (Rossi, 2013; Engel & West, 2005). On the volatility side, GARCH-class models remain foundational because they explicitly model clustering and asymmetric shock transmission (Bollerslev, 1986; Nelson, 1991). Recent applied work has expanded into boosted trees and deep learning, but systematic reviews still report instability, overfitting, and sensitivity to sample regime and feature design (Chen & Guestrin, 2016; Sezer et al., 2020). Our repository-level findings align with this literature: classical volatility structure remains useful, but direct point forecasting is fragile.

2.2 News Sentiment, GDELT, and Event-Derived Market Features

Textual information has long been used as an alternative source of financial signal. Media pessimism can forecast subsequent price pressure (Tetlock, 2007); large-scale collective mood measures can carry short-horizon market information (Bollen et al., 2011); and text mining has become a recurring theme in financial prediction research (Nassirtoussi et al., 2014; Wu et al., 2020). GDELT is especially attractive because it exposes structured event metadata, Goldstein conflict-cooperation scores, tone values, and mention counts at global scale (Leetaru & Schrod, 2013). The gap addressed here is not access to news data per se, but the need to move beyond naive aggregation. The repository shows that raw aggregate scores are weak, while theme filtering, lag design, and event-intensity engineering create more useful features.

2.3 LLMs, Multi-Agent Reasoning, and Finance

LLMs have recently been studied as financial reasoners, sentiment extractors, and agent-like forecasters (Lopez-Lira & Tang, 2023; Xie et al., 2023; Wu et al., 2023; Li et al., 2024). Prompt engineering and multi-agent decomposition are often used to diversify reasoning paths (Wei et al., 2022). Yet the application of LLMs to numerical FX forecasting remains under-specified: the main risks are anchoring on recent price, prompt bias, narrative overconfidence, and lack

Table 1: Major research tracks represented in the repository.

Track	Core Files / Folders	Primary Research Question	Saved Artifacts Used Here
Phase A/B/C	phase-a/, Phase-B/, Phase-C/	Can decomposed and thematically filtered news signal identify IMF ₃ -style volatility and danger days?	correlation_results.csv, classifier_performance.png, balanced_confusion_matrix.png, lead-lag reports
GARCH branch	GARCH/	Does EGARCH plus engineered news/macro covariates improve return and volatility modeling?	garch_comparison.csv, feature_importance.csv, EGARCH plots
Ensemble / Monte Carlo	ensemble_model/, monte_carlo_simulations/	Which quantitative family surmounts the USD/INR forecasting task, and what is the right uncertainty representation?	ultimate_model_metrics.csv, ultimate_weights.csv, Monte Carlo statistics and fan-chart outputs
Gemini personas	gemini_preds/	Can a debiased multi-persona LLM system improve short-horizon forecasts?	simulation_summary_v3.csv, simulation_summary_v4.csv, Jan 2026 live outputs
latest/	latest/	How should the older Gemini workflows be modularized into tasks, agents, cache, and meta-learning?	Treated here as an architectural bridge; the folder contains code but not a directly comparable saved plot suite
final phase	final phase/	Does a market-memory-first, walk-forward, embargoed hybrid system outperform raw feature stacks and persona-only baselines?	ablation_summary.csv, run_summary.md, multi-run comparison plots
TEPC	TEPC/	Does topology plus coupled chaos create a stronger breakout/return forecasting representation for INR/USD?	strict raw-GDELT and livepull run summaries, plot panels, persona overlay comparisons

of temporal learning. The repository addresses these issues in two ways. First, Gemini V4 introduces explicit debiasing via symmetric prompting, randomized direction order, and adaptive weighting. Second, the later **final phase** and TEPC systems reduce raw price exposure altogether by giving personas structured memory snapshots instead of unmediated spot-price context.

3 Repository-Wide Research Design

This paper is built from saved code, reports, plots, and output artifacts spanning the entire repository. Some branches are historical, others are current; some target volatility, others next-day price, and others breakout classification. Table 1 summarizes how these branches fit together.

Two methodological cautions follow from this design. First, the paper reports *branch-native* metrics rather than forcing all outputs onto an artificial unified leaderboard. Second, only reproducible artifacts present in the repository are reported numerically. The user request mentioned ARDL and DMI-style comparisons, but no versioned executable ARDL or DMI output files were present in the repository snapshot. We therefore report only implemented baselines rather than retrofitting undocumented numbers.

Table 2: Core datasets used across the repository.

Dataset	Role	Rows / Records	Date Range	Notes
india_news_gr_combined_sorted.csv	Raw India GDELT news	2,546,999	2023-01-01 to 2023-12-28	Main India event archive used for thematic filtering and TEPC raw aggregation
usa_news_combined_sorted.csv	Raw USA GDELT news	14,377,323	2023-01-01 to 2023-12-30	Cross-country comparison and TEPC raw USD-macro news nodes
usd_inr_exchange_rates_year.csv	Daily FX level	255	2024-12-30 to 2023-12-29	One-year USD/INR series used by early scripts
Phase-B/merged_training_data.csv	Thematic + IMF ₃ training frame	254	2023-01-02 to 2023-12-26	Central file for Phase-B/Phase-C GARCH-style workflows
combined_goldstein_exchange_rates.csv	Bilateral Goldstein layer	253	2023-01-02 to 2023-12-29	India/USA Goldstein summaries merged with USD/INR
political_news_exchange_merged.csv	Political news aggregate layer	252	2023-01-02 to 2023-12-24	Event count, tone dispersion, Goldstein mean/std
Super_Matrix_Dataset.csv	Long-horizon macro master frame	1,825	2019-01-02 to 2026-01-05	Main multi-source dataset for later refaction and memory-based systems
TEPC live/strict feature frames	Graph-structured node panels	214 strict rows; 321 livepull rows	strict: 2023-02-13 to 2023-12-24; livepull feature mags: 2023-01-14 to 2026-04-10	Derived inside TEPC runs rather than stored as one canonical root CSV

4 Data and Methodology

4.1 Data Layers

The repository integrates several data layers rather than a single flat table. At minimum these include daily USD/INR levels, India and USA GDELT event streams, FRED macro variables, bilateral Goldstein aggregates, thematic GDELT summaries, and later graph-structured TEPC node feeds. Table 2 lists the most important datasets used across branches.

4.2 Phase A: Variational Mode Decomposition

The first major methodological decision in the repository is to *not* predict the raw USD/INR price with news. Instead, Phase A decomposes the exchange-rate signal into lower-frequency structure and high-frequency residual noise using Variational Mode Decomposition (VMD) (Dragomiretskiy & Zosso, 2014). The intuition is simple: macro fundamentals drive the long-run trend, while news is more likely to modulate the short-horizon residual.

Let $f(t)$ denote the exchange-rate series. VMD seeks modes $u_k(t)$ and center frequencies ω_k by minimizing the total bandwidth of the demodulated modes subject to reconstruction:

$$\min_{\{u_k\}, \{\omega_k\}} \sum_{k=1}^K \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] e^{-j\omega_k t} \right\|_2^2 \quad \text{s.t.} \quad \sum_{k=1}^K u_k = f. \quad (1)$$

The repository uses $K = 3$ modes and targets IMF₃, the highest-frequency component, as the volatility/noise layer most likely to absorb daily news shocks. This decomposition step is one of the most important conceptual choices in the project: it reframes the problem away from exact long-run valuation and toward event-sensitive movement intensity.

4.3 Phase B: Thematic Filtering and Lag Engineering

Naive sentiment aggregation proved too weak for this market. The repository therefore moves to domain-driven thematic filtering, grouping India-related GDELT articles into four currency-relevant themes:

1. **Economy** (139,953 articles): RBI policy, inflation, GDP, forex, reserves, trade, taxation.
2. **Conflict** (560,938 articles): protests, crisis, border tension, attacks, strikes, military events.
3. **Policy** (594,924 articles): cabinet actions, legislation, ministerial and parliamentary policy events.
4. **Corporate** (53,779 articles): major Indian corporates, mergers, scandals, acquisitions, firm-specific shocks.

From these themes the repository constructs 16 daily features spanning tone, Goldstein, counts, and spike variables. A key engineered feature is the impact-weighted Goldstein score:

$$\text{Goldstein}_{\text{weighted}} = \sum_i \text{GoldsteinScale}_i \times \text{NumMentions}_i, \quad (2)$$

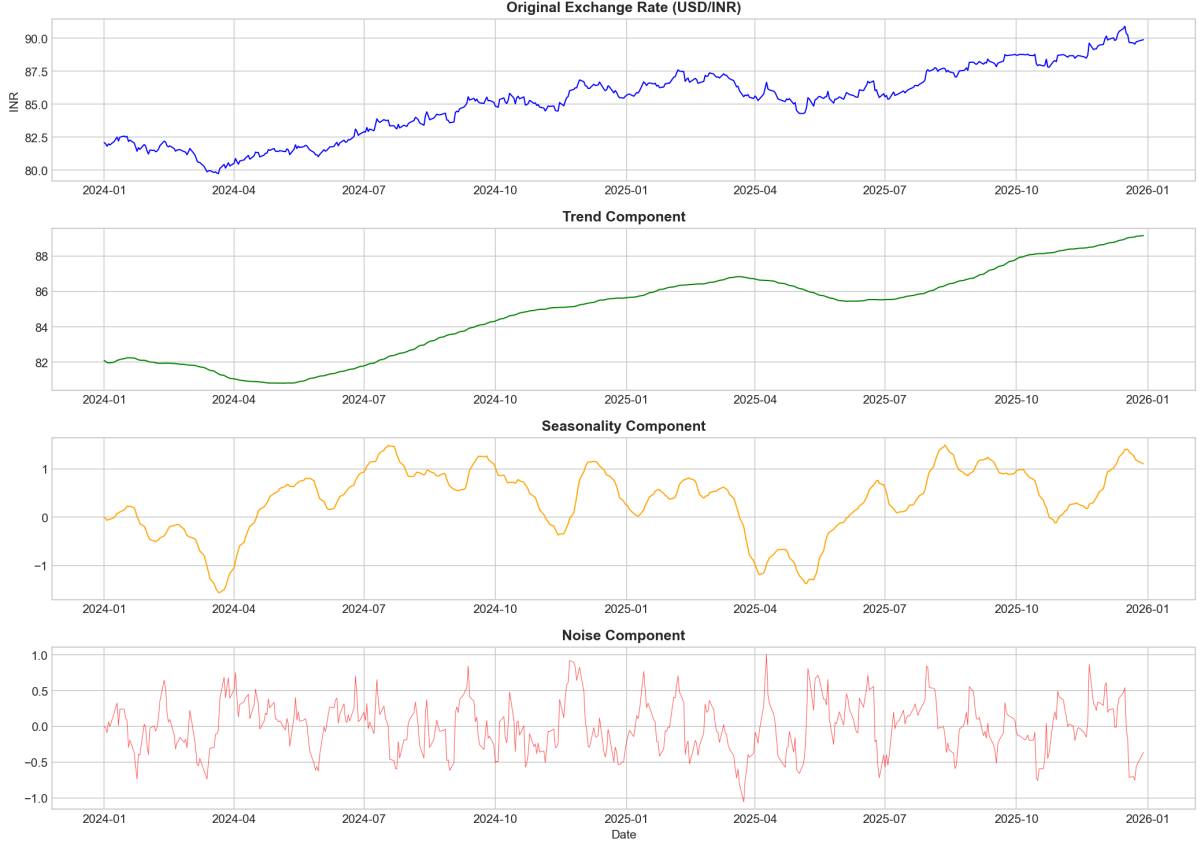


Figure 2: Representative VMD decomposition output saved by the ensemble branch. The decomposition separates the USD/INR series into trend, seasonal/cyclic, and noise-like components, with the high-frequency component treated as the most news-sensitive target in the early phases of the project.

which distinguishes between a low-attention event and a widely propagated market shock. Later branches extend the same logic into rolling averages, signed log transforms, event heat, and shared memory pressure variables.

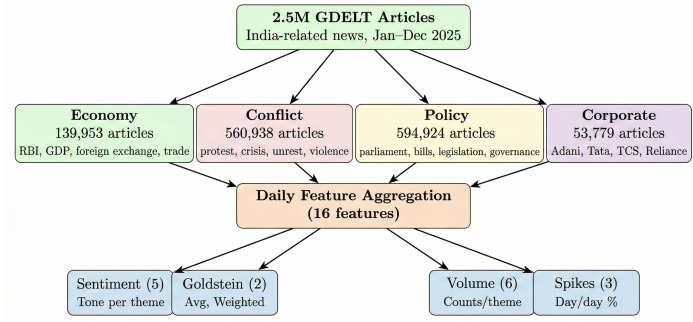
4.4 Phase C: Danger-Day Classification

Phase C formalizes a classification problem instead of a direct price-regression problem. “Danger days” are defined as the top 20% of observations by IMF_3 magnitude. For each of the 16 base features, 5 lagged variants are created, yielding an 80-dimensional feature space. The saved `danger_signal_classifier.py` implementation uses XGBoost with class-imbalance handling and an F1-optimized decision threshold.

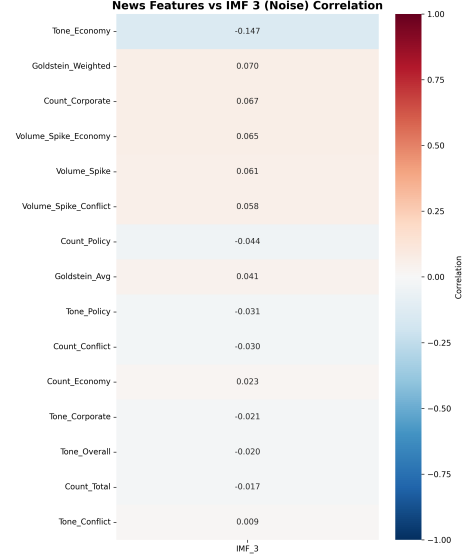
This framing is central to the paper’s argument. In a market where exact next-day price is hard to predict, a classifier that says “tomorrow is likely to be abnormal” can still be practically valuable for hedging, risk limits, and treasury execution. The repository’s downstream findings consistently support this volatility-first view.

4.5 Quantitative Forecast Families: EGARCH, Ensemble Models, and Monte Carlo

The middle repository branches explore three complementary quantitative strategies.



(a) Theme-filtering and aggregation pipeline used to turn raw GDELT into daily features.



(b) Correlation heatmap from the Phase-B outputs.

Figure 3: Phase-B transforms raw GDELT event streams into market-relevant daily features rather than using undifferentiated aggregate tone.

EGARCH and hybrid volatility modeling. The EGARCH branch models log variance to capture asymmetric shock effects:

$$\log(\sigma_t^2) = \omega + \alpha|z_{t-1}| + \gamma z_{t-1} + \beta \log(\sigma_{t-1}^2). \quad (3)$$

This branch also layers XGBoost over EGARCH-derived and news-derived covariates, allowing conditional volatility to act as a structural prior while tree models absorb nonlinear macro and sentiment interactions.

Large ensemble comparison. The ensemble branch evaluates ARIMA, SARIMA, Holt-Winters, GARCH, Monte Carlo, GDELT regressors, VMD trend models, deep learning models (LSTM, GRU, Attention), and a final optimized ensemble. This branch is valuable precisely because it documents many failures. Rather than assuming that more complexity is better, it records where model families collapse under realistic USD/INR conditions.

Monte Carlo scenario forecasting. The Monte Carlo branch shifts from point prediction to distributions. Four variants are saved: standard GBM, regime-switching, jump diffusion, and sentiment-augmented drift. This branch is best interpreted as a risk and scenario engine rather than a direct competition for next-day point accuracy.

4.6 Gemini V4: Debiased Multi-Persona LLM Forecasting

The Gemini track evolves through several versions, but V4 is the crucial one. It combines a statistical predictor with a 9-persona LLM committee and explicitly targets the failure modes seen in earlier prompting approaches: anchoring, primacy effects, and overconfident directional bias. The personas include macro, rates, flow, RBI, commodity, technical, sentiment, risk, and quant roles.

V4’s debiasing techniques are as important as the personas themselves:

1. symmetric bull and bear case presentation,

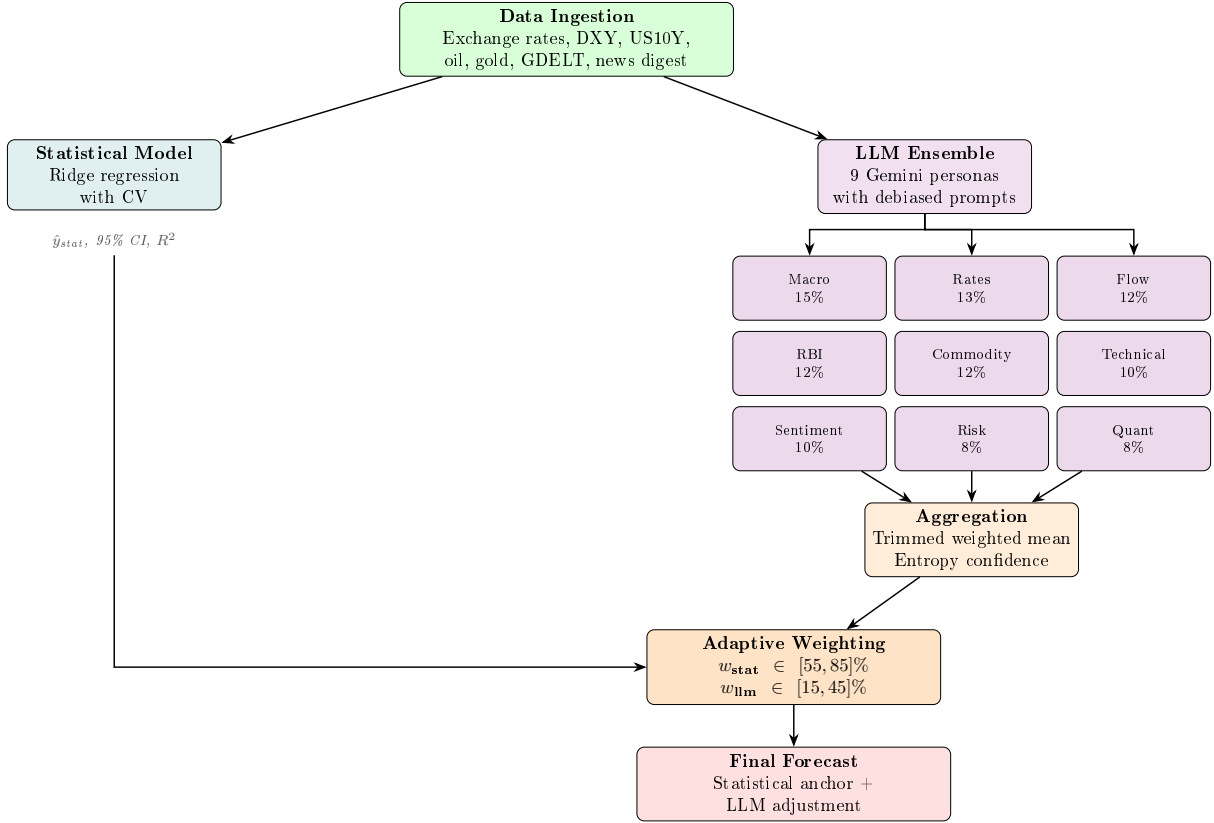


Figure 4: Gemini V4 architecture as represented in the repository’s historical LLM track. The LLM does not replace the statistical model; it perturbs and annotates it.

2. randomized direction order,
3. explicit confidence intervals and statistical model quality,
4. persona-level historical accountability,
5. bias feedback based on prior directional skew,
6. pip-bounded adjustment magnitudes.

The later repository branches inherit this lesson: persona systems are useful only when they operate as constrained overlays on top of structured quantitative state.

4.7 final phase: Market-Memory-First Hybrid Forecasting

The **final phase** system is the most current engineering-oriented branch in the repository. It departs from the earlier Gemini scripts in three ways.

First, it creates a grouped feature frame spanning technical, macro, master-sentiment, Goldstein, thematic, political, and memory features. Second, it introduces a shared market-memory snapshot that compresses recent history into stable state variables: oil pressure, dollar pressure, carry pressure, India stress, bilateral divergence, event heat, volatility stress, reversal pressure, and a regime label. Third, it evaluates everything under walk-forward backtests with an information embargo, rather than using the most recent price as an implicit shortcut. Concretely, the feature builder predicts a forward target shifted by `forecast_horizon_days + embargo_days`, so the default task becomes a two-day-gap forecast rather than a trivial persistence exercise.

The persona engine in **final phase** also differs from the older Gemini branch. Rule personas and LLM personas both consume the same structured market memory, and calibration weights

are updated from matured forecast outcomes. This makes the system auditable and branch-comparable in a way the earlier ad hoc scripts were not.

4.8 TEPC: Topology-Enabled Predictions from Chaos

TEPC is the most conceptually ambitious branch in the repository. It stops treating USD/INR as a single scalar time series and instead models it as a node in a time-varying macro-geopolitical network. The node universe includes direct market nodes such as INRUSD, DXY, BRENT, GOLD, and US10Y, plus composite nodes such as India sentiment, US sentiment, Goldstein spread, geopolitical risk, and theme pressure.

For each rolling window, TEPC computes a correlation matrix across node shock series, converts correlation into distance,

$$d_{ij} = \sqrt{\frac{1 - \rho_{ij}}{2}}, \quad (4)$$

and then applies an exponential kernel to obtain a dense adjacency matrix,

$$A_{ij} = \exp\left(-\frac{d_{ij}}{\sigma}\right), \quad A_{ii} = 0. \quad (5)$$

From the resulting Laplacian $L = D - A$, TEPC extracts degree, density, Fiedler value, spectral entropy, filtration component counts, and target-node adjacency features.

The chaos layer assigns each node a coupled Lorenz oscillator:

$$\dot{x} = \sigma(y - x) - \epsilon(Lx), \quad (6)$$

$$\dot{y} = x(\rho - z) - y - 0.35\epsilon(Ly), \quad (7)$$

$$\dot{z} = xy - \beta z - 0.15\epsilon(Lz), \quad (8)$$

with initial conditions derived deterministically from recent volatility, drift, and range position. TEPC then records synchronization, target-node phase response, finite-time Lyapunov-style divergence, and coupling response as predictive features. A strict configuration further enforces `response_lag_days = 2`, explicitly preventing access to the target day’s immediately prior price. This makes TEPC the repository’s clearest attempt to escape trivial lag-copying.

5 Experimental Setup

5.1 Branch-Specific Evaluation Protocols

Because the repository is a research program rather than one frozen benchmark, each branch uses the evaluation protocol most appropriate to its question. Table 3 summarizes the main ones used in this paper.

5.2 Implemented Baselines

The repository contains the following implemented comparison families:

- classical time-series baselines: ARIMA, SARIMA, Holt–Winters, Theta;
- volatility baselines: GARCH, GJR-GARCH, EGARCH;
- ML baselines: OLS, Ridge, Random Forest, XGBoost, Gradient Boosting;
- deep learning baselines: LSTM, GRU, Attention;
- scenario baselines: GBM, jump diffusion, sentiment-augmented drift, regime switching;

Table 3: Evaluation protocols used across major branches.

Branch	Target	Evaluation Window	Primary Metrics	Key Design Constraint
Phase C	IMF ₃ danger-day classification and 3-class direction	Chronological 80/20 split over Phase-B merged data	precision, recall, F1, accuracy, tuned threshold	Top 20% IMF ₃ defines “danger”
GARCH	Volatility fit and hybrid return modeling	Full saved branch outputs on 2025 daily data	log-likelihood, AIC, BIC, feature importance	Asymmetry and persistence are first-class outputs
Ensemble / ultimate model comparison	Next-period point forecast	Saved ensemble hold-out artifacts	RMSE, MAE, R^2 , final optimized weights	Many model families are compared under the same branch
Gemini V3/V4	Next-day price and direction	2023 full-year saved summary (260 rows) + Jan 2026 live test (21 rows)	mean absolute error %, sign accuracy, weight ranges	Statistical anchor plus persona adjustment
final phase	Future price, return, direction under walk-forward evaluation	100-day and 30-day walk-forward backtests	MAE price, RMSE price, MAE return, directional accuracy, sign-F1	Effective two-day target gap via horizon + embargo
TEPC strict raw	Breakout, forward return, near-term volatility	80-day walk-forward strict raw-GDELT run	breakout accuracy, macro-F1, MAE return, volatility MAE	Response lag = 2 days; no flat lag-copying
TEPC livepull	Same TEPC tasks on extended node feed	100-day livepull walk-forward run	breakout accuracy, macro-F1, MAE return	Optional cross-FX and live-news nodes retained when coverage permits

- LLM baselines and ablations: Gemini V3, Gemini V4, rule personas, memory-only, and later multi-provider LLM persona matrices;
- topology/chaos ablations: macro baseline, topology-only, chaos-only, topology+chaos, TEPC full, persona-only, and TEPC persona blends.

No saved ARDL or DMI benchmark outputs were found in the repository snapshot. We therefore do not fabricate ARDL/DMI rows in the result tables. Where the user requested such comparisons, we interpret the scientific intent as comparing against the strongest available implemented econometric and technical baselines.

6 Results

6.1 Raw Aggregate News is Weak; Structured and Lagged News is Better

The first clear repository-wide lesson is negative: raw aggregate news statistics are not enough. Goldstein aggregated directly against IMF₃ is almost useless; naive political-news enhancement can even worsen regression models. Only after thematic filtering, lag analysis, and stateful feature engineering does signal become usable.

This is precisely why the Indian FX market is poorly suited to simplistic direct application of many models. USD/INR does not reward a “feed in raw sentiment, predict tomorrow’s price” workflow. It requires filtering, alignment, and delayed-state representation.

6.2 Phase C: Danger-Day Classification Works Better Than Direct Price Regression

The saved Phase-C artifacts support the repository’s shift toward classification. Danger days are rare, so precision matters more than aggregate accuracy. At the F1-optimized threshold of 0.38, the classifier achieves a substantial improvement over a random 20% precision baseline.

The broader meaningful-pattern analysis also reports 41.2% accuracy on UP/DOWN/STABLE directional classification, again above the naive baseline. The core result is not that the model

Table 4: Early-stage signal discovery across the repository.

Artifact / Statistic	Value	Interpretation
Goldstein vs. IMF ₃ correlation	0.009965	Raw bilateral Goldstein aggregation alone has negligible direct relationship with the decomposed noise target.
US average tone vs. USD/INR correlation	0.1639	Cross-country tone carries weak but nonzero information; it is not sufficient as a standalone predictor.
Phase-B Tone_Economy vs. IMF ₃	-0.1466 (p = 0.0194)	Theme-filtered economic tone becomes statistically meaningful once irrelevant articles are removed.
Meaningful-pattern event count correlation with volatility	-0.1294	Event density helps characterize abnormal movement regimes.
Peak Goldstein lag	9 days	The strongest Goldstein relationship appears with delay rather than contemporaneously.
Peak tone lag	12 days	Tone shifts also appear delayed, supporting lagged rather than same-day feature design.
Political-news enhancement on RF/XGBoost	RMSE worsened by 1.18% / 1.78%	Naive injection of political features can degrade price regressors when decomposition and lag structure are absent.

Table 5: Phase-C danger-day classification results on the saved test split (73 days).

Statistic	Value
True positives	12
False positives	18
True negatives	32
False negatives	11
Precision	40%
Recall	52%
F1-score	0.45
Accuracy	60%
Optimal threshold	0.38

predicts exact prices particularly well; it does not. The result is that state-dependent volatility and directional stress are tractable.

6.3 EGARCH Remains Useful, but the Large Ensemble Exposes Many Model Failures

Table 6 shows that EGARCH is the strongest of the repository’s GARCH-family fits under the saved AIC/BIC comparison.

The hybrid branch’s top XGBoost feature importances are also informative: **Realized_Vol_20d** dominates at 27.5%, followed by **India_Avg_Goldstein** at 15.8%, **DTWEXBGS** at 9.6%, and **Goldstein_Avg** at 8.2%. This is economically plausible. Volatility structure still matters most, but bilateral event pressure and dollar conditions are meaningful secondary drivers.

The ensemble branch reinforces the same lesson even more strongly: many sophisticated models do not survive the task. Table 7 reports selected models from the saved ultimate ensemble comparison.

The optimized ensemble weights collapse to a highly interpretable solution: 95.45% **MA_Momentum** and 4.55% **GDELT_Ridge**, with effectively zero weight on the rest. This is the clearest repository-level evidence that the Indian FX market is not hospitable to many off-the-shelf direct point-prediction models. Even SARIMA, the specific baseline requested by the user, is decisively

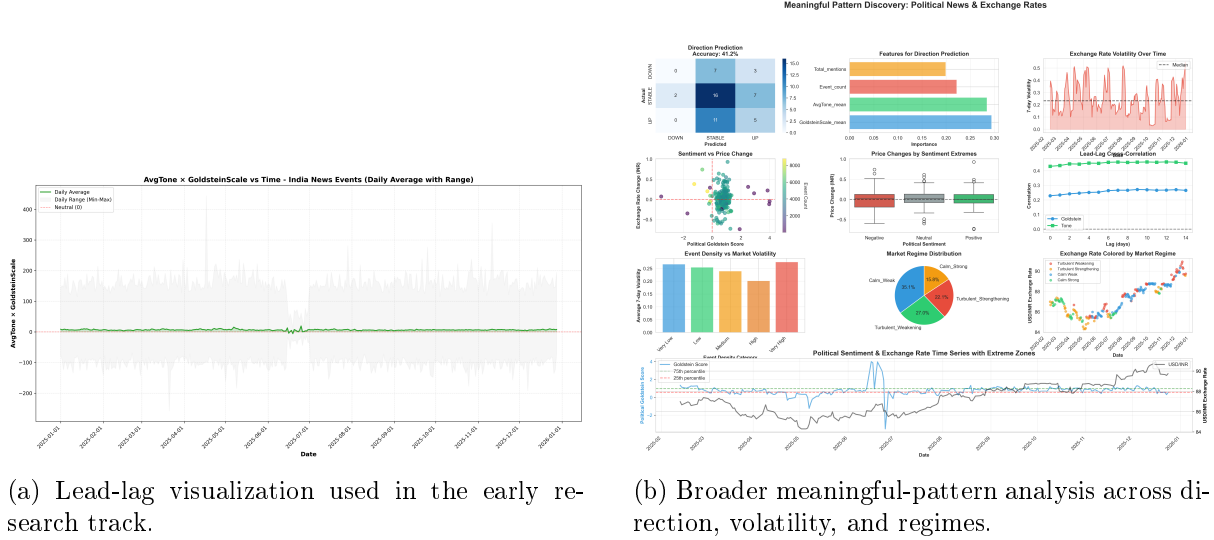


Figure 5: The strongest signal in the early repository is delayed and regime-like, not a contemporaneous one-shot news-to-price mapping.

Table 6: GARCH-family comparison from `GARCH/output/garch_comparison.csv`.

Model	Log-Lik.	AIC	BIC	Asymmetry
EGARCH	−23.166	60.331	85.037	0.2017
GARCH	−24.225	60.450	81.626	0.0000
GJR-GARCH	−23.678	61.356	86.062	−0.2519

outperformed here.

6.4 Monte Carlo is Best Used as a Distributional Risk Engine

The Monte Carlo branch is most valuable when treated as a scenario system. Table 8 summarizes the 30-day saved forecast distribution statistics.

The regime-switching specification is the most balanced saved variant: it preserves uncertainty, avoids the excessive dispersion of jump diffusion, and produces a plausible central trajectory for risk management.

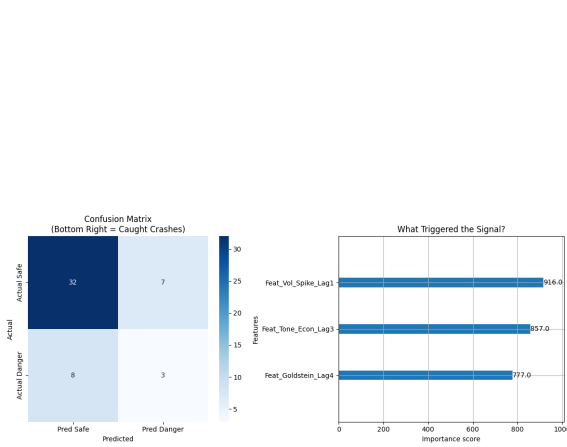
6.5 Gemini V4 Improves Over V3, but Only as a Constrained Overlay

The repository’s LLM results are strongest when the LLM is treated as a bounded adjustment layer rather than a stand-alone forecaster. Using the saved per-day summary files, Table 9 compares V3 and V4 under a sign-of-change directional evaluation.

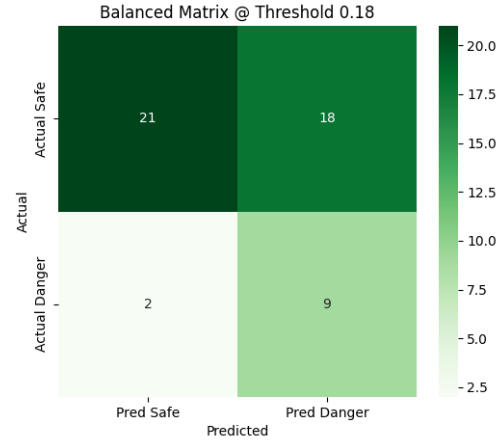
Project presentations summarize V4 directional performance more conservatively (roughly 54–58%) under a different scoring convention. The branch summary files nevertheless confirm the same qualitative result: V4 improves on V3, and the improvement is attributable to better prompt design and better adaptive weighting rather than to more aggressive LLM control.

6.6 final phase: Market Memory Beats Heavier Blends in the 100-Day Backtest

The most current hybrid branch produces one of the paper’s most surprising results. On the 100-day walk-forward backtest, the best model is not the full feature stack, nor the rule personas, nor the standalone Gemini run. It is the `memory_only_no_personas` ablation.

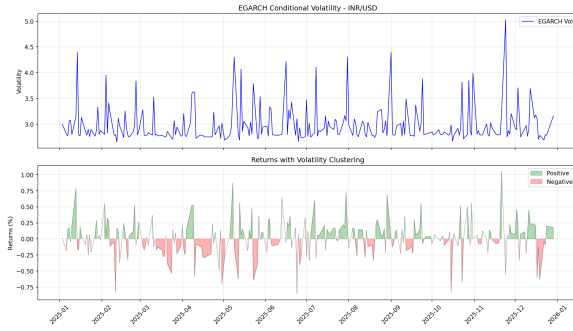


(a) Classifier performance panel.

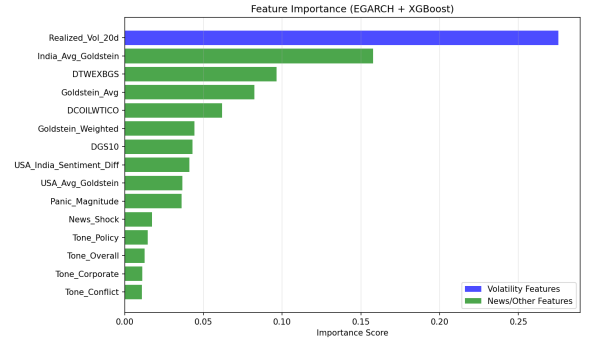


(b) Balanced confusion matrix at the tuned threshold.

Figure 6: Phase-C classification results. The practical value lies in risk flagging rather than exact valuation.



(a) EGARCH conditional volatility dynamics.



(b) Hybrid EGARCH + XGBoost feature importance.

Figure 7: The GARCH branch remains valuable as a volatility and feature-prior engine, even when direct point forecasting remains difficult.

This result suggests that, once immediate-price anchoring is removed, a compact shared-state representation can outperform a larger kitchen-sink feature stack. It also implies that the repository’s late-stage progress is not primarily about adding more raw inputs, but about learning the right latent market summary.

The 30-day LLM matrix further sharpens this result:

Here the model selection lesson is even sharper: for this short 30-day window, news-free or memory-heavy baselines beat the richer persona stacks on price MAE. The standalone 100-day Gemini 2.5 Flash backtest (`11m_gemini_100d`) still performs reasonably well at MAE price 0.2726 and directional accuracy 58%, but it does not beat the memory-only baseline.

6.7 TEPC: Chaos and Topology Become Useful Once Timing is Made Strict

The TEPC branch is where the repository most directly answers the “new generation” framing in the user’s request. It is not merely a larger feature table. It is a different forecasting ontology: USD/INR as one node in a synchronized macro system.

Under the strict raw-GDELT 80-day run with a two-day response lag, the full TEPC stack is the best experiment by the branch’s macro-F1-first ranking rule:

Table 7: Selected models from `ensemble_model/output/ultimate_model_metrics.csv`.

Model	RMSE	MAE	R^2
MA_Momentum	0.4950	0.3980	0.5711
Mini_Ensemble	0.5673	0.4442	0.4366
GARCH	0.5939	0.4594	0.3826
Monte_Carlo	0.6473	0.4898	0.2666
Theta	0.6685	0.4980	0.2178
GDELT_Ridge	0.7229	0.6270	0.0853
VMD_Trend	1.0036	0.8247	−0.7630
ARIMA	1.1505	0.8949	−1.3168
SARIMA	1.1574	0.9022	−1.3448
GRU	1.1580	0.9029	−1.3472
LSTM	1.1914	0.9282	−1.4844
Attention	2.6716	2.5081	−11.4933
Gemini_Adjusted	6.5674	6.5435	−74.4936

Table 8: Thirty-day Monte Carlo forecast statistics from `monte_carlo_statistics.csv`.

Model	Median	Std	5th Pctl.	95th Pctl.
Standard GBM	90.446	1.390	88.172	92.740
Regime-Switching	90.370	1.336	88.232	92.618
Jump Diffusion	87.260	3.940	80.714	93.610
Sentiment-Augmented	91.235	1.398	88.950	93.528

This table is revealing. Topology-only and chaos-only are individually useful but incomplete; their combination improves macro-F1 over either alone; and the full stack remains strongest overall. In other words, the branch supports the claim that topological and chaotic summaries are not gimmicks once the task is timed honestly.

The persona overlay produces a second nuance:

The rule-persona-only system attains the highest raw breakout accuracy, but the README and run notes explain why: it acts as an abstaining range predictor in a range-heavy regime. The blended TEPC persona system, by contrast, produces the best return MAE. The branch therefore separates classifier quality, regime abstention behavior, and continuous return quality rather than collapsing them into one metric.

The livepull 100-day run adds one more important result: chaos features are the strongest classifier in that extended setup. Specifically, `chaos_only` reaches 80% breakout accuracy and macro-F1 0.404, while `tepc_full` remains competitive and delivers the best return MAE at 0.003430. This is a strong indication that the chaos layer is not decorative; it becomes especially

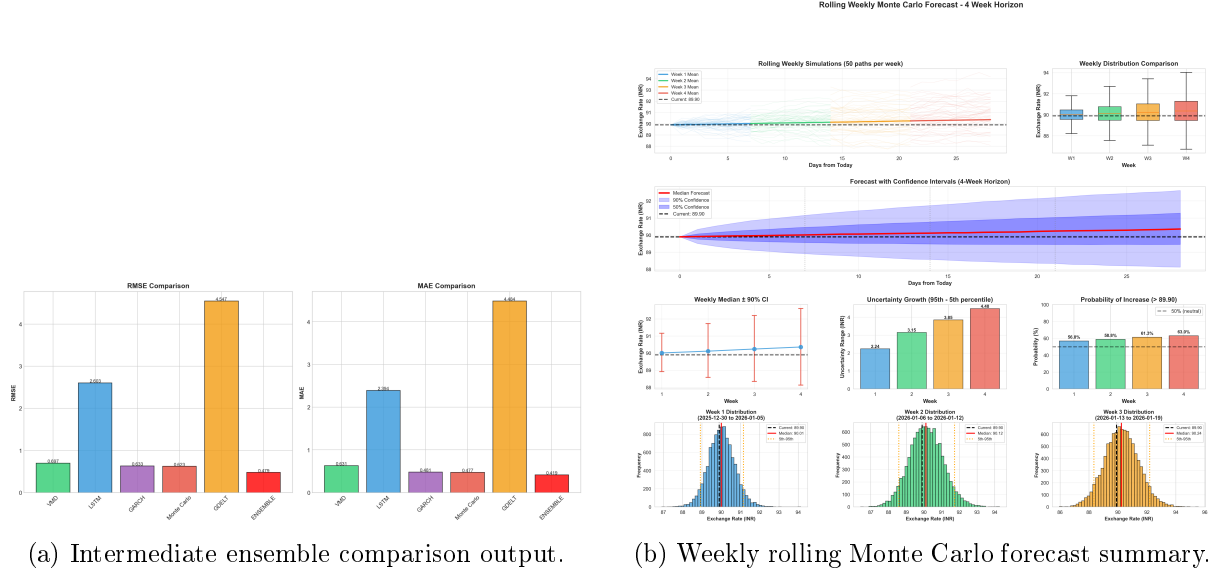


Figure 8: The quantitative middle branch of the repository ultimately favors simple trend-following plus explicit uncertainty modeling over high-capacity direct forecasters.

Table 9: Gemini V3/V4 comparison from saved daily summary files.

System	Days	Mean Abs. Error (%)	Direction Acc.	Stat / LLM Weights
V3 hybrid days	240	0.2741	61.25%	65–85 / 15–35
V4 hybrid days	240	0.2695	64.17%	55–85 / 15–45
V4 Jan 2026 live	21 total	0.3216	57.14%	62–85 / 15–38

useful once richer cross-asset node feeds are available.

6.8 Cross-Branch Synthesis

Taken together, the repository’s results support five broad conclusions.

1. USD/INR is hostile to naive direct price prediction. Negative R^2 across ARIMA, SARIMA, LSTM, GRU, Attention, several news-augmented regressors, and the older Gemini-adjusted ensemble demonstrate that more complexity is not enough. This is exactly the sense in which the Indian FX market is “not suitable” for many standard models: without strong structural constraints, they mostly learn noise, persistence, or evaluation artifacts.

2. News is useful only after structure is imposed. Raw aggregate Goldstein signal is near zero against IMF_3 , and naive political-news augmentation worsens regressors. News becomes useful after theme filtering, lag selection, event-intensity weighting, or memory/topology embedding.

3. Classification and state estimation beat raw regression. Phase C danger-day classification, final phase memory compression, and TEPC breakout forecasting are all more robust than exact one-step direct price regression.

4. LLMs help most as bounded overlays. Gemini V4 improves over V3, but the strongest later results come when LLMs or personas consume structured memory and are prevented from dominating the final forecast.

Table 10: Selected January 2026 V4 live predictions.

Date	Prediction	Actual	Error (%)	Consensus
2026-01-02	89.9981	89.9619	+0.0402	Warmup
2026-01-16	90.4564	90.3616	+0.1049	100% higher
2026-01-29	92.0165	92.0408	-0.0265	100% higher
2026-01-30	91.8205	91.7819	+0.0421	83.5% higher

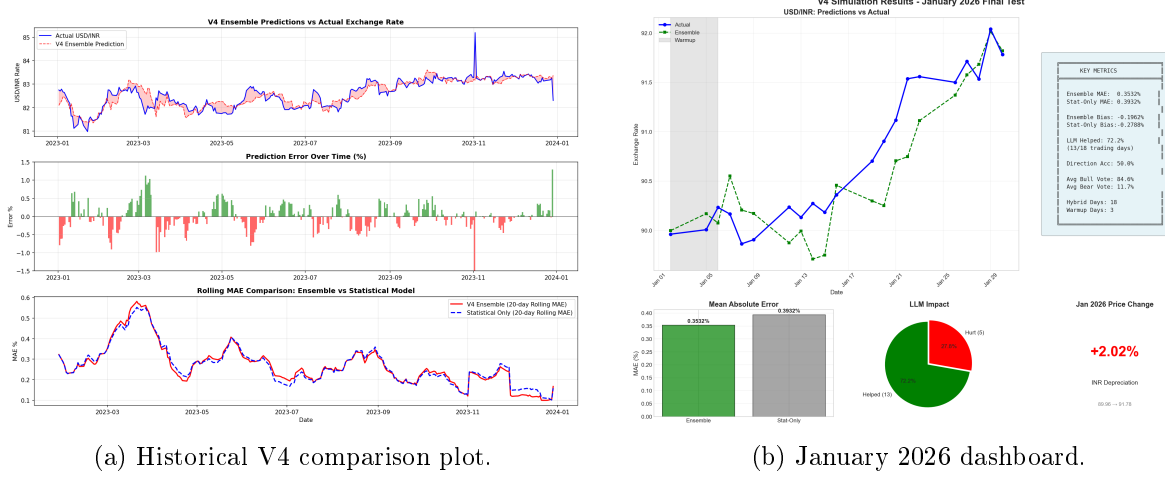


Figure 9: Gemini V4 adds value only when it is constrained by statistical anchors, explicit confidence logic, and persona diversification.

5. Topology and chaos become meaningful once timing is strict. TEPC’s strongest results appear specifically in the strict, lagged, raw-GDELT runs where trivial lag-copying is removed. This is exactly the setting in which a network/chaos representation should matter.

7 Discussion

The repository can be read as a progressive attempt to answer a single question: *what representation of the Indian FX market is stable enough to forecast?* The early answer was “not raw price, but decomposed noise and danger-day risk.” The middle answer was “not a single econometric or neural architecture, but a small combination of robust quantitative priors.” The later answer was “not raw tabular features alone, but compact shared market state.” TEPC adds a final answer: “sometimes the right object is not a scalar series at all, but a synchronized macro network.”

This progression also explains why some results appear paradoxical if read superficially. It may seem strange that memory-only outperforms a richer late-stage full model, or that chaos-only is the best TEPC classifier in one setting while TEPC full is best on macro-F1 in another. But those results are coherent once USD/INR is treated as a regime-sensitive, intervention-prone market. In such a market, extra inputs help only if they are compressed correctly; otherwise they increase variance more than signal.

The same logic resolves the role of LLMs. The repository does not support the claim that LLMs should replace classical forecasting engines. It supports a narrower, more defensible claim: multi-persona LLM systems can contribute useful incremental information when they are debiased, bounded, calibrated, and supplied with structured state rather than raw price anchors.

Table 11: **final phase** 100-day backtest ablations.

Experiment	MAE Price	RMSE Price	MAE Return	Dir. Acc.
memory_only_no_personas	0.2421	0.3290	0.002718	0.61
stat_ml_no_gdelt	0.2473	0.3359	0.002779	0.49
rule_personas	0.2901	0.4182	0.003260	0.58
stat_ml_full	0.3005	0.4361	0.003377	0.58
stat_ml_no_macro	0.3524	0.5507	0.003955	0.58

Table 12: **final phase** 30-day LLM matrix and ablation comparison.

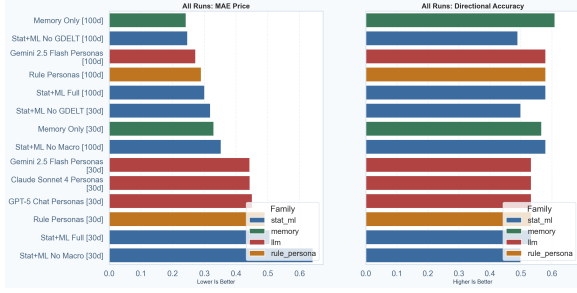
Experiment	MAE Price	RMSE Price	MAE Return	Dir. Acc.
stat_ml_no_gdelt	0.3188	0.3966	0.003540	0.5000
memory_only_no_personas	0.3295	0.3966	0.003658	0.5667
Gemini 2.5 Flash personas	0.4433	0.5696	0.004936	0.5333
Claude Sonnet 4 personas	0.4439	0.5698	0.004942	0.5333
GPT-5 Chat personas	0.4508	0.5816	0.005019	0.5333
rule_personas	0.4907	0.6355	0.005463	0.5333
stat_ml_full	0.5060	0.6578	0.005634	0.5333
stat_ml_no_macro	0.6419	0.8914	0.007142	0.5000

8 Limitations

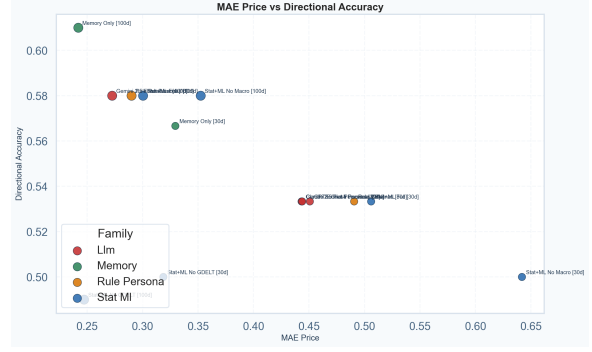
- **Heterogeneous benchmarks.** The repository contains multiple branches with different date ranges, objectives, and metrics. This paper treats them as a research program, not a single apples-to-apples benchmark.
- **Documentation drift.** Some older markdown files report slightly different summary statistics from the saved CSVs. Where this occurred, the paper prioritizes direct artifact values and explicitly notes when evaluation conventions differ.
- **Missing requested baselines.** ARDL and DMI-style benchmarks were requested conceptually, but no reproducible saved artifacts for those methods were found in the repository snapshot. They are therefore discussed only as absent baselines, not as reported empirical results.
- **Limited stable windows for some branches.** Phase-B/Phase-C and several GARCH-related analyses are effectively 2025 experiments. TEPC strict raw-GDELT runs are currently tied to local raw news coverage availability.
- **LLM reproducibility.** The Gemini and later multi-provider persona branches depend on external model APIs and prompting behavior, which are more reproducible than free-form qualitative use but still less stable than fully local numeric pipelines.
- **No transaction-cost model.** The repository studies predictability, regime detection, and risk. It does not yet include execution slippage, spread costs, or treasury workflow constraints needed for deployment.

9 Conclusion

This paper documents the full repository as a single evolving research program on USD/INR forecasting. The overarching conclusion is not that one final model “solves” FX prediction. It is that different layers of the market are predictable with different tools. The early phases



(a) Cross-run leaderboard.



(b) MAE-versus-directional-accuracy Pareto view.

Figure 10: **final phase** comparison plots. The best late-stage models are not the most complex ones; compressed memory and selective feature exclusion are often stronger.

Table 13: TEPC strict raw-GDELT 80-day ablation results.

Experiment	Breakout Acc.	Macro-F1	MAE Return	RMSE Return	MAE Volatility
tepc_full	0.7750	0.3370	0.002746	0.003765	0.000791
topology_chaos	0.6875	0.3030	0.002991	0.004136	0.000732
macro_baseline	0.7625	0.2884	0.002782	0.003755	0.000974
chaos_only	0.7375	0.2830	0.003080	0.004088	0.000873
topology_only	0.6875	0.2757	0.003352	0.004422	0.000730

show that decomposed volatility and lagged thematic sentiment contain usable delayed structure. The GARCH and ensemble branches show that many direct price models fail, and that simple momentum plus a small amount of structured signal is often stronger than heavyweight architectures. Gemini V4 demonstrates that debiased persona-based LLM forecasting can help as a bounded overlay. The **final phase** stack shows that compressed market memory can outperform richer feature blends under honest walk-forward evaluation. TEPC finally shows that topology and chaos are not cosmetic additions; under strict lagged timing they become genuinely informative breakout and regime features.

If the repository has a single scientific message, it is this: the “new generation” of prediction systems for USD/INR is not one more large model. It is a layered methodology that combines decomposition, event filtering, calibrated state representation, and network-aware dynamics, while remaining skeptical of any method that wins only by copying the last price.

A Feature and Node Inventory

A.1 Phase-B Feature Groups

The early signal pipeline uses 16 daily base features:

- tone: Tone_Economy, Tone_Conflict, Tone_Policy, Tone_Corporate, Tone_Overall;
- Goldstein: Goldstein_Avg, Goldstein_Weighted;
- counts: theme-level counts and total count;
- spikes: Volume_Spike, Volume_Spike_Economy, Volume_Spike_Conflict.

Each base feature is later lagged 1–5 days, producing an 80-dimensional feature space for Phase C.

Table 14: TEPC strict persona-v3 overlay comparison (80-day run).

Experiment	Breakout Acc.	Macro-F1	MAE Return
tepc_full	0.7750	0.3370	0.002746
persona_rule_only	0.8000	0.2963	0.002886
tepc_persona_blend	0.7875	0.2937	0.002695

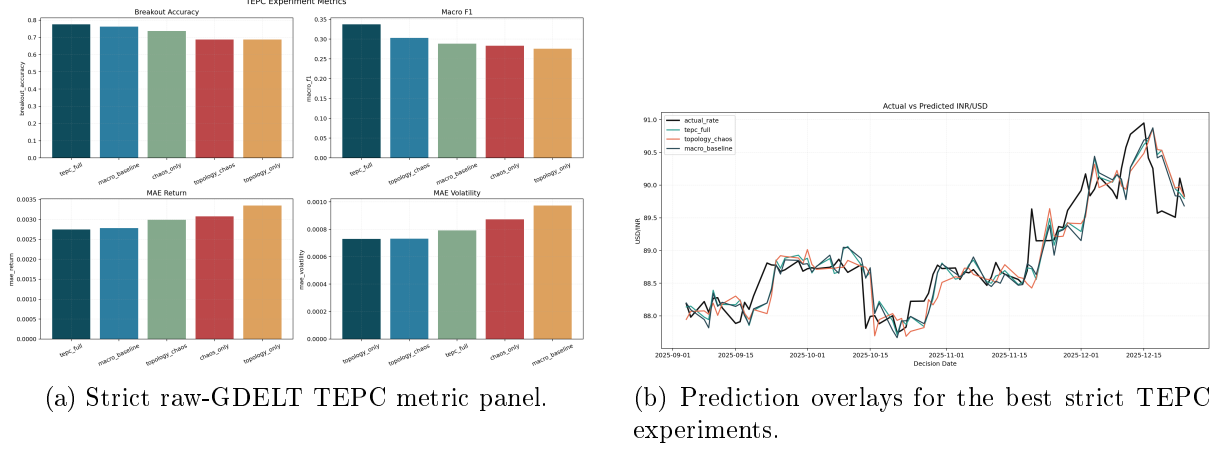


Figure 11: TEPC’s strict backtest results. The topology/chaos branch becomes useful once the task explicitly prevents lag-copying and makes regime dynamics first-class.

A.2 final phase Feature Groups

The saved feature builder in `final_phase/final_phase/features.py` constructs seven feature groups:

1. **technical**: multi-horizon returns, volatility, trend spread, z-score, range position, drawdown, momentum stability;
2. **macro**: oil, gold, US10Y, DXY returns and z-scores, plus derived pressure terms;
3. **master sentiment**: rolling India and US tone, panic, stability, mention spikes, and cross-stress variables;
4. **goldstein**: India/US Goldstein means, differences, weighted averages, event pressure;
5. **thematic**: rolling theme tone blocks, weighted Goldstein log, theme conflict spike, log total count;
6. **political**: rolling political Goldstein/tone means, article and mention intensity, event spikes;
7. **memory**: macro pressure, sentiment pressure, event heat, volatility stress, carry pressure, reversal pressure, regime score.

A.3 TEPC Node Universe

The local TEPC node universe contains direct and composite nodes:

- direct market nodes: INRUSD, DXY, BRENT, GOLD, US10Y;
- composite sentiment/geopolitical nodes: INDIA_SENTIMENT, US_SENTIMENT, GOLDSTEIN_SPREAD, GEO_RISK, THEME_PRESSURE;
- optional live or cross-FX nodes when coverage permits: EURUSD, GBPINR, USDCNH, CNHUSD, LIVE_INDIA_FX_NEWS, LIVE_USD_MACRO_NEWS, LIVE_GEO_RISK.

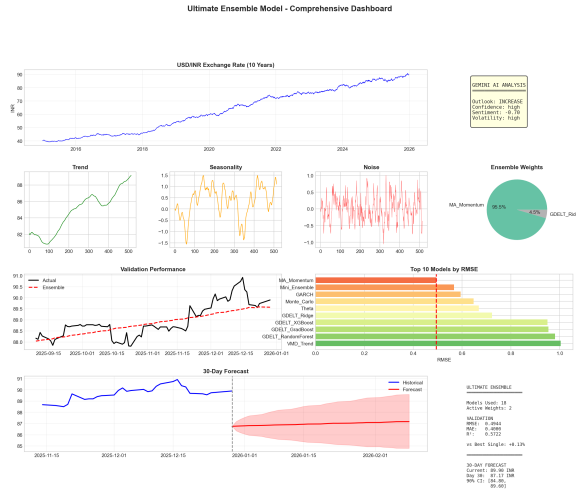
Table 15: Recommended model family by forecasting objective and horizon.

Objective / Horizon	Recommended Branch	Why
Danger-day / abnormal volatility flagging	Phase C XGBoost classifier	Best early branch for actionable risk alerts; precision and recall are more relevant than exact price fit.
Short-horizon narrative-aware next-day forecasting	Gemini V4 hybrid	Useful when a human wants a news-conditioned, persona-diversified view rather than a pure statistical forecast.
Walk-forward short-horizon forecasting under honest information embargo	<code>final phase</code> memory-only or no-GDELT ablations	Strongest late-stage MAE and directional results in the 100-day and 30-day back-tests.
Breakout classification in a coupled macro regime	TEPC chaos-heavy variants and <code>tepc_full</code>	Best repository branch for network-state and breakout interpretation under strict lagged timing.
Scenario analysis over weeks	Regime-switching Monte Carlo	Best when uncertainty bands and tail scenarios matter more than single-point prediction.
Simple practical baseline	MA_Momentum + small GDELT ridge weight	Strongest quantitative baseline in the ultimate ensemble branch and a useful reality check against over-modeling.

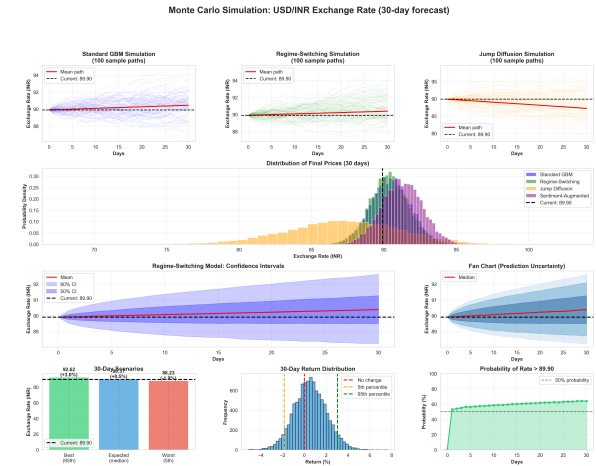
B Additional Figure Gallery

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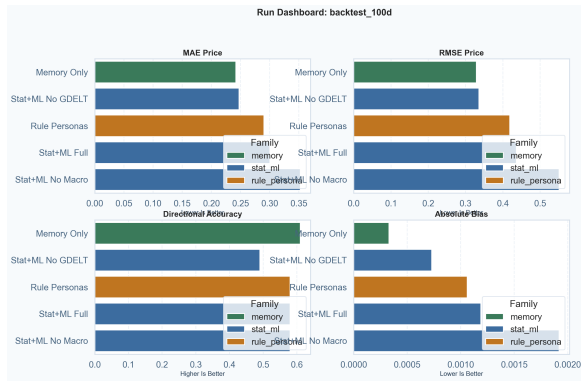


(a) Ultimate ensemble dashboard.

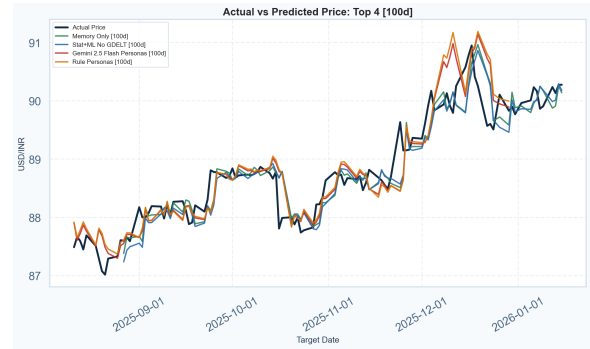


(b) Full Monte Carlo results panel.

Figure 12: Additional quantitative branch diagnostics.

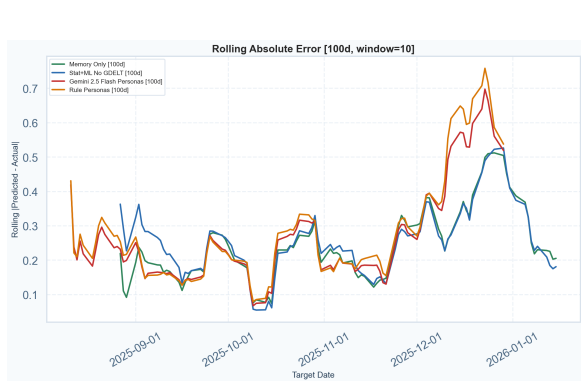


(a) final phase 100-day dashboard.

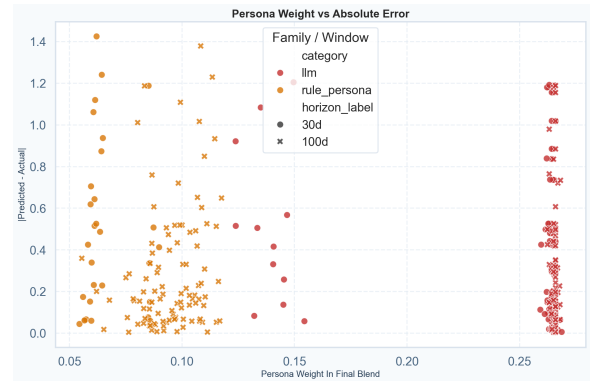


(b) Top 100-day prediction overlays.

Figure 13: Additional final phase plots.

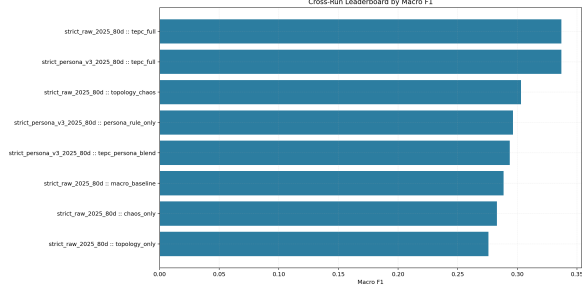


(a) Rolling absolute error over 100 days.

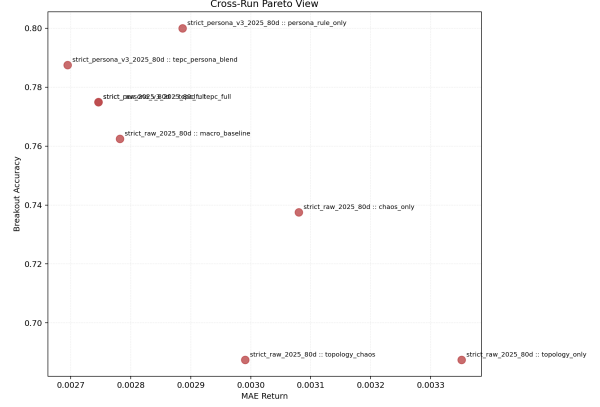


(b) Persona weight versus error relationship.

Figure 14: Late-stage diagnostics from the market-memory branch.

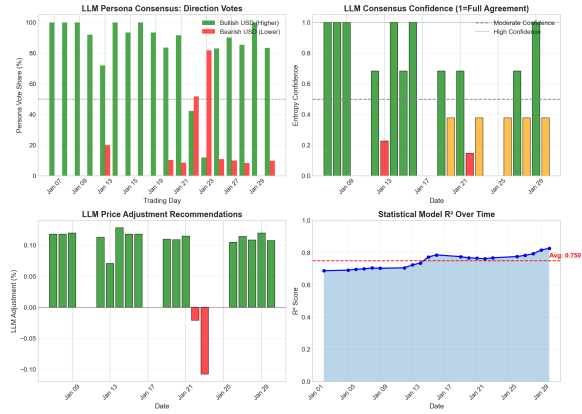


(a) TEPC cross-run leaderboard.

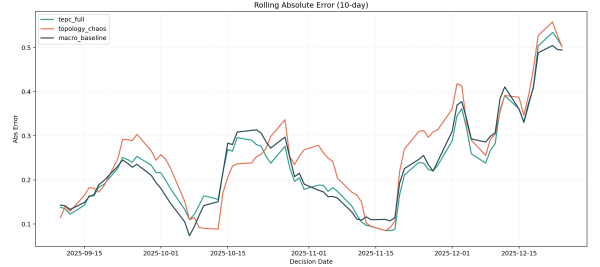


(b) TEPC cross-run Pareto comparison.

Figure 15: Additional TEPC comparison plots across strict raw and persona-overlay runs.



(a) Gemini January 2026 LLM analysis panel.



(b) TEPC strict rolling absolute error comparison.

Figure 16: Additional diagnostics for the LLM and topology-chaos branches.

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